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Inventor(s)

YEOM; Gil Choun

BATTERY PACK

Abstract

A battery pack may include battery cells; a cooling plate on one side of the battery cells; and an insulation member having an insulation part, a pair of first heat transfer parts on both sides of the insulation part, and a first layer in at least a portion of the first heat transfer part and having a higher thermal conductivity than the insulation part. A temperature deviation between top and bottom portions of an electrode plate inside a cell can be reduced, thereby improving the cooling performance of the cell. In addition, the lifespan of a cell can be extended by improving cooling performance, and the safety of a battery pack can be improved by preventing events such as deterioration and heat generation.

Inventors: YEOM; Gil Choun (Yongin-si, KR)

Applicant: Samsung SDI Co., Ltd. (Yongin-si, KR)

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

[0001] The present application claims priority to and the benefit of Korean Patent Application No. 10-2024-0030417, filed on Mar. 4, 2024, in the Korean Intellectual Property Office, the entire disclosure of which is incorporated herein by reference.

BACKGROUND

1. Field

[0002] Embodiments of the present disclosure relate to a battery pack.

2. Description of the Related Art

[0003] In general, unlike primary batteries which are not rechargeable, a secondary battery can be discharged and charged. A secondary battery is used as an energy source of, for example, a mobile device, an electric vehicle, a hybrid vehicle, an electric bicycle or an uninterruptible power supply (UPS). According to the type of external device used with the batteries, a secondary battery may be used in the form of a single battery or a battery pack in which a plurality of batteries are connected. Secondary batteries in the form of battery packs are mainly applied to devices that operate for a long time and require high power, such as hybrid cars or electric cars.

[0004] In battery packs applied to automobiles, safety is an important factor, and thus improvement of cooling performance, which is one of the greatest impacts on safety, is needed.

[0005] The above information disclosed in this Background section is only for enhancement of understanding of the background of the invention and therefore it may contain information that does not constitute prior art.

SUMMARY

[0006] Embodiments of the present disclosure provide a battery pack which can improve cooling performance and safety by reducing a temperature deviation between top and bottom portions of an electrode assembly inside a battery cell.

[0007] A battery pack according to an embodiment of the present disclosure may include: battery cells; a cooling plate on one side of the battery cells; and an insulation member between adjacent battery cells having an insulation part, a pair of first heat transfer parts on both sides of the insulation part, and a first layer in at least a portion of the first heat transfer part that has a higher thermal conductivity than the insulation part.

[0008] The insulation part may include an insulating material.

[0009] The insulation part may have a shape corresponding to the long sides of the battery cells.

[0010] The top end of the insulation part may have substantially the same height as the top ends of the battery cells or may have a height that protrudes from the top ends of the battery cells.

[0011] The pair of first heat transfer parts may be between the long side of one of the battery cells and the long side of the insulation part.

[0012] The first heat transfer part may have substantially the same height as the top end of the insulation part or may have a height lower than the top end of the insulation part.

[0013] The first heat transfer part may be a metal plate having a higher thermal conductivity than the insulation part.

[0014] The cooling plate may be at the bottom ends of the battery cells.

[0015] The first heat transfer part may include a support part in which the lower end facing the cooling plate is bent at a preset angle, and a heat conducting layer on the plate surface facing the

insulation part and having a higher thermal conductivity than the insulation part.

[0016] The support part may be bent upward at an angle of less than 90 degrees relative to the cooling plate.

[0017] The end of the support part that is bent upward may support the lower end of the insulation part.

[0018] The insulation member may further include a second heat transfer part including a first layer and a second layer in a region excluding the first layer.

[0019] The first layer and the second layer may be on the plate surface of the first heat transfer part facing one of the battery cells.

[0020] Each of the battery cells may include a hexahedral case, an electrode assembly accommodated in the case, a cap plate coupled to the top of the case, and a pair of electrode terminals electrically connected to the electrode assembly and on the cap plate.

[0021] The first layer may be in a region adjacent to the electrode terminal.

[0022] The second layer may include a material having a lower thermal conductivity than the first layer.

[0023] Each of the first layer and the second layer may be a coating layer on the first heat transfer part.

[0024] A thermal interface material (TIM) between the support part, the battery cells, and the cooling plate may be included.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0025] FIG. 1 is a perspective view showing a battery pack according to an embodiment of the present disclosure.

[0026] FIG. 2 is an exploded perspective view of the battery pack according to FIG. 1.

[0027] FIG. 3 is a diagram schematically showing cross sections of some cells and an insulation member according to FIG. 1.

[0028] FIG. 4 is an enlarged schematic diagram showing the structure of the insulation member according to FIG. 3.

[0029] FIG. 5 is a front view showing a partial structure of the insulation member according to FIG. 3.

DETAILED DESCRIPTION

[0030] Examples of the present disclosure are provided to more completely explain the present disclosure to those skilled in the art, and the following examples may be modified in various other forms. The present disclosure, however, may be embodied in many different forms and should not be construed as being limited to the example (or exemplary) embodiments set forth herein. Rather, these example embodiments are provided so that this disclosure will be thorough and complete and will convey the aspects and features of the present disclosure to those skilled in the art.

[0031] In addition, in the accompanying drawings, sizes or thicknesses of various components are exaggerated for brevity and clarity. Like numbers refer to like elements throughout. As used herein, the term “and/or” includes any and all combinations of one or more of the associated listed items. In addition, it will be understood that when an element A is referred to as being “connected to” an element B, the element A can be directly connected to the element B or an intervening element C may be present therebetween such that the element A and the element B are indirectly connected to each other.

[0032] The terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting of the disclosure. As used herein, the singular forms are intended to include the plural forms as well, unless the context clearly indicates otherwise. It will be further

understood that the terms that the terms “comprise or include” and/or “comprising or including,” when used in this specification, specify the presence of stated features, numbers, steps, operations, elements, and/or components, but do not preclude the presence or addition of one or more other features, numbers, steps, operations, elements, components, and/or groups thereof.

[0033] It will be understood that, although the terms first, second, etc. may be used herein to describe various members, elements, regions, layers and/or sections, these members, elements, regions, layers and/or sections should not be limited by these terms. These terms are only used to distinguish one member, element, region, layer and/or section from another. Thus, for example, a first member, a first element, a first region, a first layer and/or a first section discussed below could be termed a second member, a second element, a second region, a second layer and/or a second section without departing from the teachings of the present disclosure.

[0034] Spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper,” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if the element or feature in the figures is turned over, elements described as “below” or “beneath” other elements or features would then be oriented “on” or “above” the other elements or features. Thus, the exemplary term “below” can encompass both an orientation of above and below.

[0035] Hereinafter, a battery pack according to an embodiment of the present disclosure will be described in detail with reference to the accompanying drawings.

[0036] FIG. 1 is a perspective view showing a battery pack **10** according to an embodiment of the present disclosure. FIG. 2 is an exploded perspective view of the battery pack **10** according to FIG. 1.

[0037] Referring to FIGS. 1 and 2, the battery pack **10** may include a plurality of battery cells **100**, busbars **200** electrically connecting the battery cells **100**, an end plate **300** and a side plate **400** for supporting and fixing the battery cells **100**, and an insulation plate **500** for insulating the battery cells **100** from the end plate **300**. In addition, the battery pack **10** may further include a plurality of insulation members **600** between the battery cells **100** and a cooling plate **700** below the battery cells **100**.

[0038] In one or more embodiments, each of the battery cells **100** may be a lithium-ion secondary battery with a rectangular parallelepiped shape. In one or more embodiments, each of the battery cells **100** has a rectangular parallelepiped shape, and the plurality of battery cells **100** may be arranged in a row along a certain direction (e.g., in the x-axis direction in FIG. 1). The plurality of battery cells **100** may be arranged so that relatively large plate surfaces thereof face each other. The insulation members **600**, which will be described later, may be between adjacent battery cells **100** to prevent (or at least mitigate) heat propagation.

[0039] Each of the battery cells **100** may include a case **110** in the shape of a rectangular parallelepiped with one side open, an electrode assembly accommodated inside the case **110**, and a cap plate **120** coupled to one open side of the case **110** to seal the case **110**. An electrode terminal **122** electrically connected to the electrode assembly may be on the cap plate **120**.

[0040] In one or more embodiments, the case **110** has an open upper surface so that the cap plate **120** can be coupled thereto. The case **110** includes a pair of long sides **112** having a relatively large area, a pair of short sides **114** connecting the long sides **112** and having a relatively small area, and a bottom surface **116** that is a short side surface and forms a bottom of the case **110**. The plurality of battery cells **100** are arranged so that long sides **112** of the case **110** face each other. The case **110** may be made of metal such as aluminum or stainless steel.

[0041] The electrode assembly is accommodated in the case **110** with an electrolyte (liquid, gel, or solid electrolyte). The electrolyte may be composed of a lithium salt such as LiPF₆ or LiBF₄ with an organic solvent such as ethylene carbonate (EC), propylene carbonate (PC),

diethyl carbonate (DEC), ethylmethyl carbonate (EMC), or dimethyl carbonate (DMC).

[0042] In one or more embodiments, the electrode assembly may be configured such that a positive electrode plate and a negative electrode plate are wound or stacked with a separator therebetween. The positive electrode plate and the negative electrode plate may include active material regions having a positive electrode active material and a negative electrode active material each coated on a thin metal foil substrate, and uncoated regions (non-coating regions) without an active material coated thereon. In one or more embodiments, the positive electrode plate may be provided by coating an active material, such as a transition metal oxide, on a substrate provided with a metal foil, such as aluminum foil. In addition, the negative electrode plate may be provided by coating an active material such as carbon, graphite, etc. on a substrate provided with a metal foil such as copper or nickel foil. The positive electrode uncoated portion and the negative electrode uncoated portion may be arranged in opposite directions. A positive electrode tab may be connected to the positive electrode uncoated portion, and a negative electrode tab may be connected to the negative electrode uncoated portion. The positive electrode tab and the negative electrode tab may be electrically connected to the electrode terminals **122** of the positive electrode and the negative electrode, respectively. The busbars **200** may be connected to the electrode terminals **122**.

[0043] The plurality of busbars **200** may electrically connect adjacent electrode terminals **122** of the plurality of battery cells **100** in series or parallel. The busbars **200** may be mounted on upper portions of the electrode terminals **122** by a busbar holder. The busbars **200** may serve to electrically connect the electrode terminals **122** to the outside.

[0044] The end plate **300** includes a pair of end plates that each may have an approximately rectangular parallelepiped shape. The end plates **300** are arranged to face the outermost ones of the battery cells **100**. In one or more embodiments, the pair of end plates **300** are arranged to be symmetrical with each other. The end plate **300** may support the plurality of battery cells **100** in the X-axis direction. The side plates **400** may be coupled to both sides of the end plates **300**. The insulation plate **500** may be between the end plate **300** and the outermost battery cell **100**.

[0045] The side plate **400** may be sized to cover at least portions of the side surfaces of the plurality of battery cells **100**. The side plate **400** may have a bar or plate shape. The side plate **400** may be arranged in a direction in which a pair of end plates **300** are connected (e.g., the side plates **400** may extend in the X-axis direction). In one or more embodiments, a pair of side plates **400**, each having a bar shape, are on each side of the end plate **300** in the y-axis direction (e.g., a first pair of side plates **400** on one side the end plates **300** and a second pair of side plates **400** on another side of the end plates **300**). First ends of the pair of side plates **400** may be coupled to the end plate **300** at one end, and second ends of the pair of side plates **400** may be coupled to the end plate **300** of the other end. First ends of the pair of side plates **400** may be coupled to the end plate **300** at one end, and second ends of the pair of side plates **400** may be coupled to the end plate **300** at the other end. In one or more embodiments, two pairs of side plates **400** are arranged to be symmetrical (or substantially symmetrical) to each other with the end plate **300** and the battery cells **100** between the two pairs of side plates **400**. Therefore, the side plates **400** support the battery cells **100** from the sides. The plurality of battery cells **100** may be physically fastened together by the end plates **300** and the side plates **400**. In addition, the side plates **400** may be made of an insulating material. In one or more embodiments in which the side plates **400** are made of a metal material, an insulation tape, etc. may be attached thereto to prevent the side plates **400** from being in contact with the battery cells **100** or being electrically conducted.

[0046] The insulation plate **500** may be between the end plate **300** and the outermost battery cell **100** adjacent to the end plate **300**. The insulation plate **500** may insulate the battery cells **100** from the end plate **300**. The insulation plate **500** may be made of a material configured to provide electrical insulation, prevent (or at least mitigate) heat transfer, and absorb shocks. In one or more embodiments, the insulation plate **500** may not be provided and the end plate **300** may replace the function of the insulation plate **500** without separately providing the insulation plate **500**.

[0047] The insulation members **600** and the cooling plate **700** may be provided to cool the plurality of battery cells **100**, as described above.

[0048] Each of the insulation members **600** is between adjacent ones of the plurality of battery cells **100**, and the cooling plate **700** is below the battery cells **100**. The insulation member will be described later.

[0049] The cooling plate **700** is in close contact with the bottom surfaces of the battery cells **100** (see **116** of FIG. **3**). Therefore, the cooling plate **700** may be sized and shaped to at least entirely cover the battery cells **100**. In one or more embodiments, the cooling plate **700** may be an aluminum alloy plate having high thermal conductivity. Therefore, the heat generated in the battery cells **100** may be released to the outside by the cooling plate **700**.

[0050] Additionally, a thermal interface material (TIM) **800** may be between the cooling plate **700** and the bottom surface **116** of each of the battery cells **100**. For convenience, FIG. **2** shows the TIM **800** provided similar to a solid-type plate, but the TIM **800** may be applied to the upper surface of the cooling plate **700**. In other words, the TIM **800** may have the same form as a coating layer coated on the top surface of the cooling plate **700**. The TIM **800** is configured to improve heat transfer performance so that the heat generated from the battery cells **100** can be more easily dissipated through the cooling plate **700**. Some of the heat generated from the battery cells **100** is transferred to the cooling plate **700** through the TIM **800**. In addition, some of the heat generated from the battery cells **100** may be transferred to the cooling plate **700** through the insulation member **600**. However, only the cooling plate **700** may be provided without the TIM **800**.

[0051] Hereinafter, the insulation member **600** will be described in detail with reference to FIGS. **3** to **5**.

[0052] FIG. **3** is a diagram schematically showing cross sections of some cells **100** and an insulation member **600** according to FIG. **1**. FIG. **4** is an enlarged schematic diagram showing the structure of the insulation member **600** according to FIG. **3**. FIG. **5** is a front view showing a partial structure of the insulation member **600** according to FIG. **3**. FIG. **4** shows various components spaced apart from each other to show the structure of the insulation member **600**. As shown in FIG. **3**, the insulation member **600** may include an insulation part **610**, a first heat transfer part **620**, and a second heat transfer part **630** in close contact with each other. In addition, to show the movement of heat, the battery cells **100** and the insulation member **600** are shown as being spaced apart from each other, but the insulation member **600** may be arranged to be in close contact with the battery cells **100**. In one or more embodiments, the insulation member **600** and the battery cells **100** may be spaced apart from each other, and a gap between the insulation member **600** and the battery cells **100** may correspond to a gap sufficient to allow for the movement of heat therebetween.

[0053] The insulation member **600** may be a substantially square plate, as shown in FIG. **2**, when viewed from the outside. The insulation member **600** may include an insulation part **610** on the innermost side, a pair of first heat transfer parts **620** on the outside of the insulation part **610**, and a second heat transfer part **630** on the outside of each of the pair of first heat transfer parts **620**.

[0054] The insulation part **610** is a rectangular plate with a predetermined thickness and may include an insulating material. The insulation part **610** is configured to block heat transfer between neighboring battery cells **100**. The top end of the insulation part **610** may have the same height (or substantially the same height) as or a greater height than the top ends of the battery cells **100**. The top ends of the first heat transfer part **620** and the second heat transfer part **630**, which will later be described, may also have the same heights (or substantially the same heights) as the top end of the insulation part **610**. In addition, the top ends of the first heat transfer part **620** and the second heat transfer part **630** may also have a smaller height (lower height) than the top end of the insulation part **610**. Therefore, at least the top end of the insulation part **610** may be at the same height (or substantially the same height) as the top ends of the battery cells **100** or may protrude further upward from the top ends of the battery cells **100** (see region A in FIGS. **3** and **4**). In addition, the insulation part **610** is configured not to be in direct contact with the cooling plate **700**. In one or

more embodiments, the first heat transfer part **620** has a structure in which the insulation part **610** and the cooling plate **700** are not in direct contact with each other.

[0055] In one or more embodiments, the first heat transfer part **620** is a pair of first heat transfer parts **620** between the long side surface **112** of the battery cells **100** and the long side surface of the insulation part **610**. That is, the pair of first heat transfer parts **620** are not provided as one body, but are two separate bodies on both long sides of the insulation part **610** (see {circle around (1)} and {circle around (2)} in FIG. 4). Therefore, the insulation part **610** has a structure in which each of the pair of long sides **112** is not direct in contact with the battery cells **100** due to the pair of first heat transfer parts **620**. When viewed from the long sides **112** of the battery cells **100** (on the basis of the x-axis direction in FIG. 2), the first heat transfer part **620** may be square (or substantially square). The first heat transfer part **620** may be a metal plate made of aluminum or other metal having high thermal conductivity. When viewed from the short sides **114** of the battery cells **100** (on the basis of the y-axis direction in FIG. 2), the lower end of the first heat transfer part **620** is bent at a predetermined angle to support the insulation part **610**. The bent lower end is referred to as the support part **622** (or support portion). A heat conducting layer **624** may be provided on a plate surface of each of the first heat transfer parts **620** facing the insulation part **610**.

[0056] The support part **622** may be bent at an angle of less than 90 degrees (e.g., the interior angle between the support part **622** and the plate surface of the first heat transfer part **620** may be less than 90 degrees). In one or more embodiments, the first heat transfer part **620** is perpendicular (or substantially perpendicular) to the cooling plate **700**, and thus if the support part **622** were bent at 90 degrees there is a possibility that the insulation part **610** may come into contact with the cooling plate **700**. Since the insulation part **610** is configured to impede heat transfer, it is desirable that the cooling plate **700** and TIM **800** not be in contact with the insulation part **610** as much as possible. The TIM **800** is applied to the upper surface of the cooling plate **700** and comes into contact with portions of the bottom and side surfaces of the battery cells **100**. Therefore, by bending the support part **622** upward at an angle of less than 90 degrees to support the insulation part **610**, it is possible to prevent (or at least mitigate) the insulation part **610** from contacting the TIM **800**, which is in contact with the cooling plate **700**. Accordingly, the support part **622** may be bent such that the insulation part **610** may not come into contact with the TIM **800**. This configuration is configured to prevent interference with the first heat transfer part **620** as much as possible when the insulation part **610** contracts or expands due to swelling of the battery cells **100**. In addition, the structure of the support part **622** can maximize (or at least increase) contact areas with the first heat transfer part **620** and the TIM **800**. That is, in an embodiment in which the first heat transfer part **620** includes the support part **622**, compared to when the first heat transfer part **620** is arranged perpendicular to the cooling plate **700**, the support part **622** can increase the contact areas with the first heat transfer part **620** and the TIM **800**. Therefore, the heat transferred through the first heat transfer part **620** can be quickly transferred to the cooling plate **700** through the TIM **800**. In order to maximize (or at least increase) the heat transfer performance, the heat conducting layer **624** is provided on the plate surface of the first heat transfer part **620** that faces the insulation part **610**. In addition, the second heat transfer part **630** may be provided on the plate surface of the first heat transfer part **620** that faces the battery cells **100** and faces away from the insulation part **610**.

[0057] The conducting heat layer **624** may be provided by applying a material having high thermal conductivity to the first heat transfer part **620** or by applying or coating a thermal interface material.

[0058] The second heat transfer part **630** may be a coating layer on the surface of the first heat transfer part **620** facing the battery cells **100**. In one or more embodiments, the second heat transfer part **630** may be an adhesive sheet on the surface of the first heat transfer part **620** facing the battery cells **100**. In one or more embodiments, the second heat transfer part **630** may include a first layer **632** and a second layer **634**.

[0059] The first layer **632** of the second heat transfer part **630** may be located in a position

corresponding to the upper region on the basis of the battery cells **100** in FIG. 3. The first layer **632** has a relatively high thermal conductivity and may include a material capable of efficiently transferring heat. The lower ends of the battery cells **100** are in contact with the TIM **800** and the cooling plate **700**, and thus, in response to heat being generated from the battery cells **100**, the temperature of the upper region of the battery cells **100** becomes higher than the lower region thereof due to heat conduction. Here, the upper and lower regions of the battery cells **100** should be understood as relative positions, not as absolute positions, according to temperatures. The area of the region being at a higher temperature than the lower region can be experimentally confirmed. The sizes of the first layer **632** and the second layer **634** may be determined accordingly. In one or more embodiments, the first layer **632** may be in a region corresponding to approximately $\frac{1}{4}$ to approximately $\frac{1}{3}$ of the total height of the battery cells **100**. In one or more embodiments, the second layer **634** may be provided in the remaining regions that are not provided with the first layer **632**. The first layer **632** is configured to receive the heat from heat-generated portions of the battery cells **100** being at relatively high temperatures. The transferred heat is transferred to the first heat transfer part **620** through the first layer **632** and to the support part **622** along the heat conducting layer **624**. Since the support part **622** is in contact with the TIM **800** and the cooling plate **700**, the heat generated from the heat-generated portions of the battery cells **100** can be quickly transferred to cooling plate **700**.

[0060] The second layer **634** may be in a position corresponding to the lower region on the basis of the battery cells **100** in FIG. 3. The second layer **634** may include a material configured to block heat. Since the lower ends of the battery cells **100** are in contact with the TIM **800** and the cooling plate **700**, some of the heat generated from the battery cells **100** is moved directly to the cooling plate **700**. In addition, since the second layer **634** blocks heat, some of the heat generated from the battery cells **100** is not conducted toward the second layer **634** and moves to the upper ends of the battery cells **100**. Therefore, the heat conducted to the upper region of the battery cells **100** can be transferred to the cooling plate **700** through the first layer **632**, as described above.

[0061] As described above, the heat generated from the battery cells **100** can be primarily cooled by the TIM **800** and the cooling plate **700** which are in direct contact with the lower portions of the battery cells **100**. Additionally, in one or more embodiments, the movement of heat to the lower side of the battery cells **100** where cooling is performed is blocked (or at least reduced) by the second layer **634**. In addition, the heat that has moved, by heat conduction, to the upper side of the battery cells **100**, is transferred to the first layer **632** and transferred to the TIM **800** and the cooling plate **700** through the first heat transfer parts **620** (see region B in FIG. 4), to then be secondarily cooled. That is, while heat transfer between neighboring battery cells **100** is blocked by the insulation member **600**, direct cooling of the battery cells **100** by the cooling plate **700** and indirect cooling of the battery cells **100** by the insulation member **600** are achieved. Accordingly, the high-temperature heat-generated portions inside the battery cells **100** can be cooled quickly, thereby reducing temperature deviations inside the battery cells **100** and improving the cooling of the battery cells **100**.

[0062] In addition, the lifespan of the battery cells **100** can be extended by improving cooling performance, and the safety of a battery pack can be improved by preventing an event such as deterioration or heat generation.

[0063] According to an embodiment of the present disclosure, a temperature deviation between top and bottom portions of an electrode plate inside a cell can be reduced, thereby improving the cooling performance of the cell.

[0064] In addition, the lifespan of a cell can be extended by improving cooling performance, and the safety of a battery pack can be improved by preventing events such as deterioration and heat generation.

[0065] While the foregoing embodiment is only one embodiment for carrying out the present disclosure, which is not limited to the embodiment, it will be understood by a person skilled in the

art that various changes in form and details may be made therein without departing from the spirit and scope of the present disclosure as defined by the following claims.

Claims

1 what is claimed is:

- 1.** A battery pack comprising: a plurality of battery cells; a cooling plate on one side of the plurality of battery cells; and an insulation member between adjacent battery cells of the plurality of battery cells, the insulation member comprising an insulation part, a pair of first heat transfer parts on both sides of the insulation part, and a first layer on at least a portion of each of the pair of first heat transfer parts, the first layer having a higher thermal conductivity than the insulation part.
- 2.** The battery pack as claimed in claim 1, wherein the insulation part comprises an insulating material.
- 3.** The battery pack as claimed in claim 2, wherein the insulation part has a shape corresponding to long sides of the plurality of battery cells.
- 4.** The battery pack as claimed in claim 3, wherein a top end of the insulation part has substantially a same height as top ends of the plurality of battery cells or wherein the insulation part has a height that protrudes from the top ends of the plurality of battery cells.
- 5.** The battery pack as claimed in claim 4, wherein each of the pair of first heat transfer parts is between a long side of one of the adjacent battery cells and a long side of the insulation part.
- 6.** The battery pack as claimed in claim 5, wherein each of the pair of first heat transfer parts has substantially a same height as a top end of the insulation part or wherein each of the pair of first heat transfer parts has a height lower than the top end of the insulation part.
- 7.** The battery pack as claimed in claim 6, wherein each of the pair of first heat transfer parts is a metal plate having a higher thermal conductivity than the insulation part.
- 8.** The battery pack as claimed in claim 7, wherein the cooling plate is at bottom ends of the plurality of battery cells.
- 9.** The battery pack as claimed in claim 8, wherein each of the pair of first heat transfer parts comprises a support part having a lower end facing the cooling plate that is bent at a preset angle, and a heat conducting layer on a plate surface facing the insulation part and having a higher thermal conductivity than the insulation part.
- 10.** The battery pack as claimed in claim 9, wherein the support part is bent upward at an angle of less than 90 degrees relative to the cooling plate.
- 11.** The battery pack as claimed in claim 10, wherein the lower end of the support part that is bent upward supports a lower end of the insulation part.
- 12.** The battery pack as claimed in claim 10, wherein the insulation member further comprises a second heat transfer part including a first layer and a second layer in a region excluding the first layer.
- 13.** The battery pack as claimed in claim 12, wherein the first layer and the second layer are on the plate surface of each of the pair of first heat transfer parts facing one of the adjacent battery cells.
- 14.** The battery pack as claimed in claim 13, wherein each of the plurality of battery cells includes a hexahedral case, an electrode assembly accommodated in the hexahedral case, a cap plate coupled to a top of the hexahedral case, and a pair of electrode terminals electrically connected to the electrode assembly and on the cap plate.
- 15.** The battery pack as claimed in claim 14, wherein the first layer is in a region adjacent to the pair of electrode terminals.
- 16.** The battery pack as claimed in claim 15, wherein the second layer comprises a material having a lower thermal conductivity than the first layer.
- 17.** The battery pack as claimed in claim 16, wherein each of the first layer and the second layer is a coating layer on each of the pair of first heat transfer parts.

18. The battery pack as claimed in claim 11, further comprising a thermal interface material (TIM) between the support part, the plurality of battery cells, and the cooling plate.

19. The battery pack as claimed in claim 17, further comprising a thermal interface material (TIM) between the support part, the plurality of battery cells, and the cooling plate.
